

UWB-Anchored LiDAR-Aided SLAM for Compute-Efficient Multi-Agent Construction Mapping

Spatiotemporal Semantic Search via 2D Detection Lifted into 3D

Priyadarshan Narayanswamy, Aryan Tajne, Dahong Luo, Parth Dua, Nishkal Hundia

Abstract—Construction environments are dynamic, visually repetitive, and often GPS-denied, which makes globally consistent 3D reconstruction difficult to maintain with vision-only SLAM over long trajectories. While LiDAR and RGB-D sensing improve local geometric fidelity and metric scale, they do not by themselves prevent global drift or simplify multi-agent map alignment at site scale. We propose a UWB-anchored, LiDAR-aided SLAM architecture tailored to construction workflows: LiDAR provides locally accurate geometry, and sparse UWB ranging constraints stabilize global alignment and reduce dependence on expensive multi-agent, loop-closure-driven map merging. Beyond geometry, we introduce a spatiotemporal semantic layer that enables semantic search over space–time by lifting 2D detections (YOLO26) into 3D using per-frame depth and ray projection, supporting queries such as “what changed where and when” and “which hazards occurred near a zone.” We prototype the system using a mobile RGB-D rig (iPhone Pro LiDAR) and a UWB anchor device for controlled evaluation, with intended deployment as a helmet-mounted depth+video+UWB tag pipeline feeding an edge server. This paper details the method, implementation, and evaluation plan; experimental results are left as placeholders for insertion.

Index Terms—Construction robotics, SLAM, RGB-D, LiDAR, UWB, multi-agent mapping, semantic mapping, spatiotemporal search, digital twin.

I. INTRODUCTION

Accurate, up-to-date 3D understanding is a prerequisite for construction progress tracking, safety monitoring, and as-built versus as-planned verification [1], [2]. However, construction sites routinely challenge traditional visual SLAM: repetitive corridor structure, low texture, dynamic occlusions, and lighting changes can degrade tracking and yield inconsistent reconstructions [3], [4]. Commodity LiDAR-enabled mobile devices have made RGB-D data collection widely accessible [5]–[7]. Yet, even with depth sensing, global drift and multi-session/multi-agent alignment remain difficult without strong global constraints.

Multi-agent SLAM promises scalability by distributing coverage across workers, but practical deployments are constrained by (i) the need for overlap to obtain inter-agent loop closures and consistent merges and (ii) the compute/communication burden of global optimization and map fusion [8]–[10]. Our thesis is that sparse UWB constraints—already used in indoor positioning systems and supported by modern standards ecosystems [11]–[13]—can serve as lightweight global priors that reduce drift and lessen the dependence on heavy loop-closure-centric merging.

Contributions. This paper makes three contributions:

- **UWB-anchored LiDAR-aided SLAM for construction.** We formulate a factor-graph view where UWB range constraints complement ORB-SLAM3 pose estimates and LiDAR geometry to stabilize global alignment [14], [15].
- **Compute-efficient multi-agent alignment with reduced overlap dependence.** We describe how anchoring to a site frame via UWB reduces reliance on crossing paths to align maps, mitigating a core operational constraint highlighted in collaborative SLAM surveys [8].
- **Spatiotemporal semantic search in 3D.** We lift 2D detections (YOLO26) into 3D using depth + ray projection to generate a semantic index over space and time, enabling construction-centric queries and change analytics [16]–[18].

II. BACKGROUND AND RELATED WORK

A. Visual and visual-inertial SLAM

Modern slam systems are commonly structured into a front-end (data association) and a back-end (MAP estimation), where loop closures are essential for bounding drift over time [3]. ORB-SLAM3 is a widely used open-source system supporting monocular, stereo, and RGB-D configurations, including multi-map capabilities [14].

B. LiDAR and RGB-D SLAM

Depth sensing reduces scale ambiguity and strengthens geometric constraints, improving local reconstruction quality relative to vision-only pipelines. LOAM demonstrates real-time lidar odometry and mapping, while RTAB-Map provides long-term, large-scale SLAM with memory management and supports lidar/visual sensing [19], [20]. Mobile LiDAR devices and AR pipelines provide dense depth and surfaces at consumer scale [5], [7].

C. UWB ranging, standards, and error modes

UWB is widely used for fine-grained ranging and indoor positioning [11]. Interoperability has been driven by IEEE 802.15.4z and industry ecosystems such as FiRa [12], [13]. In practice, UWB performance can degrade under NLoS and human-body shadowing; mitigation strategies are an active research topic [21].

D. UWB-aided odometry and construction relevance

Range-aided VIO/SLAM integrates UWB constraints to eliminate or reduce drift while within anchor coverage [15]. In construction contexts, UWB+VSLAM fusion is explicitly motivated by low texture and occlusions [4]. Recent construction-focused work demonstrates factor-graph integration of intermittent UWB with VIO, including interpolation for asynchronous range measurements [22].

E. Multi-agent SLAM and map merging complexity

Collaborative SLAM requires communication, inter-agent association, and global consistency maintenance, and it introduces resource management and robustness challenges at scale [8]. Fully distributed SLAM using constrained factor graphs (DDF-SAM) formalizes scalable computation and communication in multi-robot mapping [9]. State-of-the-art multi-robot systems such as Kimera-Multi combine distributed loop closure detection with pose graph optimization and mesh deformation, emphasizing robustness but still relying on loop closures and non-trivial back-ends [10].

III. PROBLEM STATEMENT

We aim to construct a *globally consistent* 3D map of a construction environment from mobile wearable sensing, support multiple agents without forcing overlap-heavy trajectories, and attach a spatiotemporal semantic index for retrieval and change analytics.

Let $\mathbf{x}_k \in SE(3)$ be the pose of an agent at keyframe k in its local SLAM frame. Vision (and LiDAR depth) provide relative constraints between poses, while UWB provides sparse absolute or anchor-relative constraints. We formulate estimation as a maximum a posteriori (MAP) problem on a factor graph [3]:

$$\begin{aligned} \mathbf{X}^* = \arg \min_{\mathbf{X}} & \sum_{(i,j) \in \mathcal{E}_{\text{vis}}} \|\mathbf{r}_{\text{vis}}(\mathbf{x}_i, \mathbf{x}_j)\|_{\Sigma_{\text{vis}}}^2 \\ & + \sum_{k \in \mathcal{K}_{\text{depth}}} \|\mathbf{r}_{\text{depth}}(\mathbf{x}_k)\|_{\Sigma_{\text{depth}}}^2 \\ & + \lambda \sum_{m \in \mathcal{M}_{\text{uwb}}} \|\mathbf{r}_{\text{uwb}}(\mathbf{x}(t_m))\|_{\Sigma_{\text{uwb}}}^2, \end{aligned} \quad (1)$$

where $\mathbf{r}_{\text{uwb}}$ encodes anchor range (and optionally bearing) measurements. The parameter λ balances UWB priors with SLAM constraints.

IV. METHOD

A. Sensor and deployment model

Intended deployment. Each worker wears a helmet rig with (i) forward-facing video, (ii) local depth/LiDAR (for geometry and metric scale), and (iii) a UWB tag used for site-frame anchoring. Commodity mobile LiDAR has made RGB-D collection broadly feasible [5], [7]. UWB provides accurate short-range ranging in indoor environments under favorable conditions [11].

Experimental prototype. For controlled prototyping, we use two mobile RGB-D devices (iPhone Pro LiDAR) as agents and a third device acting as a UWB anchor during

hallway traversals. Nearby Interaction provides streamed distance/direction between opted-in UWB-equipped devices [23], [24]. We emphasize that the *hardware platform is a prototype*; the method is described in platform-agnostic terms.

B. Baselines

1) *SLAM-only (ORB-SLAM3)*: We run ORB-SLAM3 as a feature-based baseline under standard configurations [14]. This baseline is expected to incur drift and reconstruction inconsistency in repetitive corridors and dynamic occlusions [3].

2) *SLAM + LiDAR*: We integrate depth to improve local geometric fidelity and scale. ORB-SLAM3 supports RGB-D configurations [14]; more generally, RGB-D and lidar SLAM systems demonstrate improved geometry constraints relative to monocular pipelines [19], [20].

C. Our method: SLAM + LiDAR + UWB constraints

The method adds sparse UWB constraints as global priors to stabilize the site frame. Following construction-focused fusion approaches, we treat UWB as asynchronous measurements and associate each range $d(t)$ with an interpolated pose $\mathbf{x}(t)$ [22].

1) *UWB ranging model*: A representative anchor ranging model (with potential antenna offset $\mathbf{o}$ in the body frame) is:

$$d(t) = \|\mathbf{p}(t) + \mathbf{R}(t)\mathbf{o} - \mathbf{a}_i\|_2 + b_i + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \quad (2)$$

which matches recent construction-oriented factor-graph formulations [22]. In practice, NLoS and human-body shadowing can bias measurements; robustification and error modeling are necessary [21].

2) *Fusion strategy*: We target a compute-efficient fusion design:

- **Front-end**: ORB-SLAM3 produces high-rate odometry/keyframes [14].
- **Geometry support**: depth aids local structure (reducing warping and scale error) [7].
- **Backend**: pose-graph / fixed-lag smoothing integrates range factors (and optional bearing when available) similarly to UWB-aided VIO literature [15].

The system outline is described in

D. Multi-agent alignment with reduced overlap dependence

Classic multi-agent SLAM often requires inter-agent loop closures to merge coordinate frames, which operationally implies agents must traverse overlap regions [8]. Our method reduces this dependence: each agent is pulled into a shared site frame by UWB anchor constraints, so global alignment does not rely solely on crossing paths. We stress *reduction*, not *elimination*: overlap still improves map quality and semantic association, but UWB provides a consistent *initialization and stabilization mechanism* for alignment.

From an estimation standpoint, each agent has a local SLAM frame W_i and a site frame S . If $\mathbf{T}_{W_i \leftarrow D_i}(t)$ is the device pose from SLAM and $\mathbf{T}_{S \leftarrow D_i}(t)$ is the UWB-anchored pose estimate, then:

$$\mathbf{T}_{S \leftarrow W_i}(t) = \mathbf{T}_{S \leftarrow D_i}(t) (\mathbf{T}_{W_i \leftarrow D_i}(t))^{-1}. \quad (3)$$

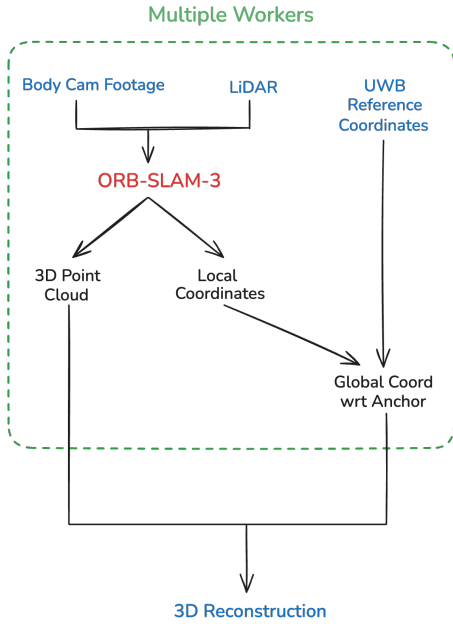


Fig. 1. *

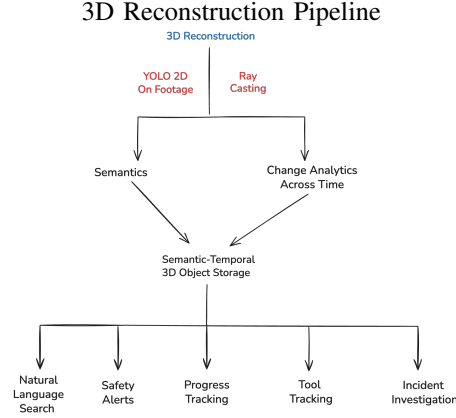


Fig. 2. *

Spatial Temporal Understanding Pipeline

Fig. 3. System Flowchart

E. Semantic layer: spatiotemporal semantic mapping and search

We aim to answer not only *what is the geometry*, but *what is where and when*. Semantic mapping systems show that fusing per-frame semantics with depth yields richer, queryable maps [16].

1) *System Overview*: We present a vision-based pipeline for semantically labeled 3D object localization that fuses monocular SLAM, object detection, point cloud association, and global positioning. Given a monocular RGB video stream, the system estimates camera poses and sparse scene geometry using ORB-SLAM3, detects objects in the image plane using YOLO26n, lifts 2D detections into 3D space via ray–point cloud intersection, and converts local SLAM coordinates into a global reference frame using an Ultra-Wideband (UWB)

positioning system.

2) *Camera Pose and Point Cloud Estimation*: For each time step t , ORB-SLAM3 provides the camera pose relative to the SLAM coordinate frame:

$$\mathbf{T}_c^s(t) = \begin{bmatrix} \mathbf{R}_c^s(t) & \mathbf{t}_c^s(t) \\ \mathbf{0}^\top & 1 \end{bmatrix} \in SE(3), \quad (4)$$

where $\mathbf{R}_c^s \in SO(3)$ and $\mathbf{t}_c^s \in \mathbb{R}^3$ denote the camera rotation and translation, respectively. In addition, ORB-SLAM3 reconstructs a sparse point cloud:

$$\mathcal{P}^s(t) = \{\mathbf{p}_i^s \in \mathbb{R}^3\}_{i=1}^{N_t}, \quad (5)$$

expressed in the SLAM coordinate frame.

3) *2D Object Detection*: Object detection is performed on the RGB image I_t using YOLO26n, producing a set of bounding boxes:

$$\mathcal{B}_t = \{(\mathbf{b}_k, c_k)\}_{k=1}^{K_t}, \quad (6)$$

where $\mathbf{b}_k = (u_{\min}, v_{\min}, u_{\max}, v_{\max})$ defines the pixel coordinates of the k -th bounding box, and c_k denotes its semantic class label.

4) *Ray Casting and 3D Object Localization*: For each bounding box $\mathbf{b}_k$, we compute a representative pixel location, defined as the box center:

$$(u_k, v_k) = \left(\frac{u_{\min} + u_{\max}}{2}, \frac{v_{\min} + v_{\max}}{2} \right). \quad (7)$$

Using the camera intrinsic matrix $\mathbf{K}$, the corresponding normalized image ray in the camera frame is obtained as:

$$\mathbf{r}_k^c = \mathbf{K}^{-1} \begin{bmatrix} u_k \\ v_k \\ 1 \end{bmatrix}, \quad \mathbf{r}_k^c \leftarrow \frac{\mathbf{r}_k^c}{\|\mathbf{r}_k^c\|}. \quad (8)$$

This ray is transformed into the SLAM coordinate frame:

$$\mathbf{r}_k^s = \mathbf{R}_c^s(t) \mathbf{r}_k^c. \quad (9)$$

We define the ray originating from the camera center $\mathbf{t}_c^s(t)$ as:

$$\mathbf{x}_k(\lambda) = \mathbf{t}_c^s(t) + \lambda \mathbf{r}_k^s, \quad \lambda > 0. \quad (10)$$

The 3D object position is estimated by finding the nearest point in the SLAM point cloud that lies along this ray:

$$\mathbf{p}_k^s = \arg \min_{\mathbf{p}_i^s \in \mathcal{P}^s(t)} \|(\mathbf{p}_i^s - \mathbf{t}_c^s) \times \mathbf{r}_k^s\|, \quad (11)$$

subject to:

$$(\mathbf{p}_i^s - \mathbf{t}_c^s) \cdot \mathbf{r}_k^s > 0. \quad (12)$$

This yields a 3D estimate $\mathbf{p}_k^s = (x_k^s, y_k^s, z_k^s)$ of the object in the SLAM coordinate frame.

5) *Global Coordinate Transformation Using UWB*: To obtain globally consistent object positions, we transform SLAM-frame coordinates into a global reference frame using UWB-based localization. Let:

$$\mathbf{T}_s^g = \begin{bmatrix} \mathbf{R}_s^g & \mathbf{t}_s^g \\ \mathbf{0}^\top & 1 \end{bmatrix} \quad (13)$$

denote the transformation from the SLAM frame to the global frame, estimated from UWB measurements.

The global object position is then computed as:

$$\begin{bmatrix} \mathbf{p}_k^g \\ 1 \end{bmatrix} = \mathbf{T}_s^g \begin{bmatrix} \mathbf{p}_k^s \\ 1 \end{bmatrix}, \quad \mathbf{p}_k^g \in \mathbb{R}^3. \quad (14)$$

6) *Spatiotemporal Semantic Object Storage*: Each detected object observation is stored as a spatiotemporal semantic tuple that jointly encodes its time of observation, semantic category, and global 3D position:

$$\mathcal{O}_k(t) = (t_k, c_k, \mathbf{p}_k^g), \quad (15)$$

where t_k denotes the timestamp of the observation, c_k is the object category predicted by YOLO26n, and $\mathbf{p}_k^g \in \mathbb{R}^3$ is the corresponding global 3D position obtained after SLAM-UWB transformation. Accumulating these tuples over time yields a persistent, semantically annotated 3D object map with temporal context.

To enable efficient spatiotemporal querying and object-level reasoning, we maintain a semantic index defined as:

$$\mathcal{I} : (t, c, \mathbf{p}) \mapsto \{\text{track_id}, \alpha, \mathcal{F}\}, \quad (16)$$

where *track_id* uniquely identifies an object instance across time, α denotes an aggregated confidence score, and $\mathcal{F}$ is the set of supporting frames in which the object was observed. The index supports object association across multiple observations and enables temporal consistency checks.

This spatiotemporal semantic representation supports a range of product-oriented queries, including:

- *Change detection*: identifying objects whose global displacement exceeds a threshold X between times t_1 and t_2 ;
- *Safety analysis*: detecting hazardous object classes within a distance X of a predefined zone Z during a specified time interval;
- *Semantic retrieval*: retrieving image frames in which objects of class c appear within a spatial neighborhood of location $\mathbf{p}$.

By explicitly coupling semantics, geometry, and time, the proposed storage and indexing scheme enables scalable construction progress monitoring, safety auditing, and spatiotemporal change detection, demonstrating the practical value of unified 3D spatial semantics for real-world analytics. [1], [2].

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Edge-server split

Following collaborative SLAM resource constraints, a practical architecture offloads heavy compute (semantic inference, fusion optimization, visualization) to an edge server while

maintaining light capture on the wearable [8]. Our prototype uses:

- **Wearable/agent (helmet rig)**: capture RGB, depth, and UWB measurements; run lightweight SLAM front-end.
- **Laptop edge server**: ingests pose/depth streams, runs optimization, semantic inference, and visualization. This avoids transmitting raw video continuously when not necessary and supports batch processing of recorded footage.

B. Data logging and synchronization

We timestamp all streams and log:

- SLAM poses per keyframe (timestamp, pose, covariances if available),
- UWB updates (timestamp, anchor id, range and optional direction, quality flags),
- semantic detections (timestamp, class, mask/box, confidence),
- point clouds/mesh snapshots (keyframe id).

Implementation note: we recommend line-delimited JSON (JSONL) for append-only logging and reproducibility; tables and plots can be generated from these logs in post-processing.

VI. PROTOTYPE VALIDATION AND QUALITATIVE RESULTS

We validated the final system in three hallway sessions using iPhone LiDAR (RGB-D) and UWB anchoring. In the final pipeline, session poses were placed in a shared global frame using UWB-derived constraints/transform estimates, then fused with SLAM trajectories for map-level consistency.

A. What Was Demonstrated

The end-to-end stack worked in the target flow:

- 1) collect RGB-D + SLAM trajectory + UWB measurements per session,
- 2) estimate/apply global alignment from UWB constraints,
- 3) generate globally positioned frame records for visualization and retrieval,
- 4) render the merged map and session trajectories in one world frame,
- 5) click a 3D location and retrieve the corresponding frame/time observation in the frontend with NLP supported queries.

B. Interactive Spatiotemporal Retrieval and NLP Queries

To validate the utility of the globally anchored map and the spatiotemporal semantic index, we developed an interactive web frontend linking the 3D geometry, trajectory data, and visual evidence. The system demonstrated the following capabilities:

- **Cross-Session Spatial Retrieval**: Users successfully queried locations within the fused global point cloud to instantly retrieve the corresponding RGB frames and timestamps across three independent mapping sessions (s_1, s_2, s_3). This confirms the efficacy of the UWB constraints in pulling local trajectories into a unified, queryable site frame.

- **Semantic Scene Analysis:** By integrating a Vision-Language Model (VLM) with the spatial index, the system successfully answered natural language queries regarding specific map coordinates. For example, querying a map junction yielded detailed architectural descriptions (e.g., identifying polished concrete, exposed cable trays, and partition walls) alongside relative spatial context.
- **4D Change Detection:** Utilizing the 3D-lifted YOLO detections, the system visualized an “evolving environment” timeline. The prototype successfully tracked the introduction and movement of objects (e.g., chairs) over a 10-minute interval.
- **Spatially-Bounded Temporal Queries:** When prompted with “how did this area change over time?”, the AI assistant accurately cross-referenced the semantic index across time steps. Crucially, the system demonstrated spatial awareness: it successfully reported the appearance and 3D coordinates of new objects when queried near their location, while correctly reporting “no change” when queried in undisturbed zones of the same map.

C. Observed Output Artifacts

From the processed run, the frontend index produced 146 valid frame-world records across 3 sessions (34, 53, 59), with one record filtered out by tracking-state gating. Timestamp association tolerance was set to 0.06 s.

D. Scope of This Hackathon Evaluation

This work demonstrates functional system integration and qualitative map/trajectory consistency under UWB anchoring. We do not claim benchmark-level quantitative performance in this submission (e.g., Chamfer distance, ATE/RPE, ICP residual statistics, runtime-memory profiling), and leave those measurements as planned follow-up experiments when time is not a constraint.

VII. APPLICATIONS: PRODUCT-ORIENTED VALUE IN CONSTRUCTION

We emphasize that improved 3D reconstruction is not only a visualization feature; it is the substrate for measurable, actionable workflows:

- **Progress and as-built updates:** integrate point clouds with BIM for discrepancy detection and progress quantification [1].
- **Safety monitoring:** detect and localize hazards; enable spatial queries (“hazards within X meters”) enabled by 3D-lifted semantics [16].
- **Change detection over time:** semantic-aided change detection methods demonstrate that semantics help interpret geometric change [2].
- **Semantic search in video+space:** workers, managers, and QA can retrieve evidence by querying object presence near a location/time window, reducing manual review.

At a system level, UWB anchoring supports stable spatiotemporal indexing: the same physical area maps to the same site frame across sessions, enabling time-series analytics.

VIII. LIMITATIONS AND FAILURE MODES

UWB degradation. NLoS, multipath, and human body shadowing can bias ranges and reduce reliability [11], [21]. Robust estimators and quality gating are required.

Infrastructure dependence. UWB constraints require anchors or an anchor device; anchor placement and coverage geometry affect outcomes [12], [13].

Asynchrony. UWB updates are not synchronized to camera frames; correct time alignment and pose interpolation is needed [22].

Semantic model domain shift. Object detectors may underperform on construction-specific imagery without adaptation; YOLO26 supports edge deployment but still depends on training/data quality [17], [25].

Privacy and governance. Continuous video capture raises privacy/security concerns; practical deployments require on-device redaction policies and access control.

IX. CONCLUSION AND FUTURE WORK

We proposed a UWB-anchored, LiDAR-aided SLAM system for construction that targets globally consistent mapping, reduced overlap dependence in multi-agent alignment, and spatiotemporal semantic search by lifting 2D detections into 3D. Future work includes robust NLoS-aware UWB error models, formal ablation on larger datasets, scaling to more agents/anchors, and tight integration between semantic tracks and factor-graph optimization.

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